

Medical Statistics - Answers lab 10

Integrated exercises with the Cleveland heart disease dataset

Part A: ANCOVA

A1. Research question

The research question is whether resting blood pressure (`trestbps`) differs between males and females after adjusting for age.

In model notation, this is assessed with:

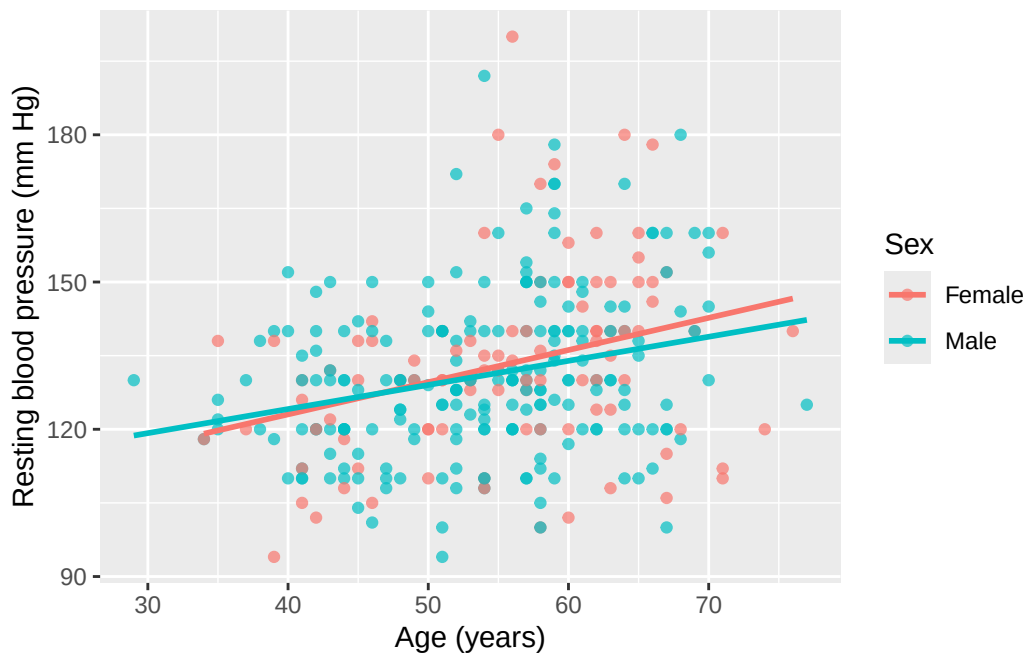
$$\text{trestbps} = \beta_0 + \beta_1 \times \text{age} + \beta_2 \times \text{sex} + \varepsilon$$

The coefficient for `sex` estimates the adjusted mean difference in resting blood pressure between males and females at the same age.

A2. Exploratory analysis

```
# A tibble: 2 x 7
  sex      n mean_trestbps sd_trestbps median_trestbps    q1    q3
  <fct> <int>      <dbl>      <dbl>          <dbl> <dbl> <dbl>
1 Female   97      133.        19.4           132   120   140
2 Male   206      131.        16.7           130   120   140
```

The mean resting blood pressure is about 133 mm Hg for females and 131 mm Hg for males. The distributions are quite similar, with both groups having a median close to 130-132 mm Hg and an interquartile range of approximately 120 to 140 mm Hg.



The scatterplot suggests a weak positive association between age and resting blood pressure. There is no strong visual evidence of a clearly non-linear relationship, so a linear adjustment for age is reasonable for this exercise.

A3. Additive ANCOVA model

Call:

```
lm(formula = trestbps ~ age + sex, data = heart_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-38.125	-11.038	-0.838	10.587	66.507

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	102.8155	6.2686	16.402	< 2e-16 ***
age	0.5478	0.1082	5.063	7.21e-07 ***
sexMale	-1.3940	2.0928	-0.666	0.506

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 16.91 on 300 degrees of freedom

Multiple R-squared: 0.08255, Adjusted R-squared: 0.07643
 F-statistic: 13.5 on 2 and 300 DF, p-value: 2.439e-06

	2.5 %	97.5 %
(Intercept)	90.4794612	115.151637
age	0.3348881	0.760724
sexMale	-5.5123753	2.724295

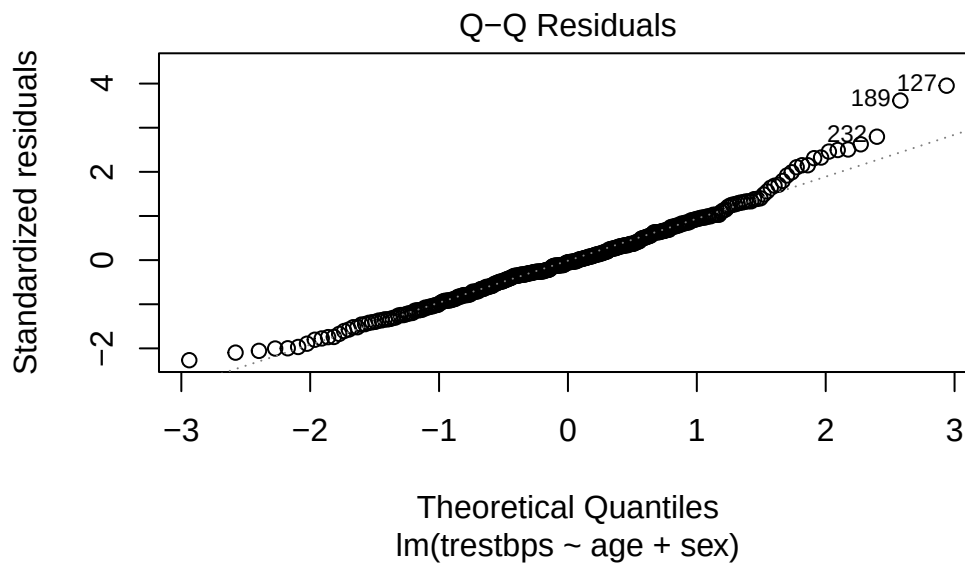
Answer

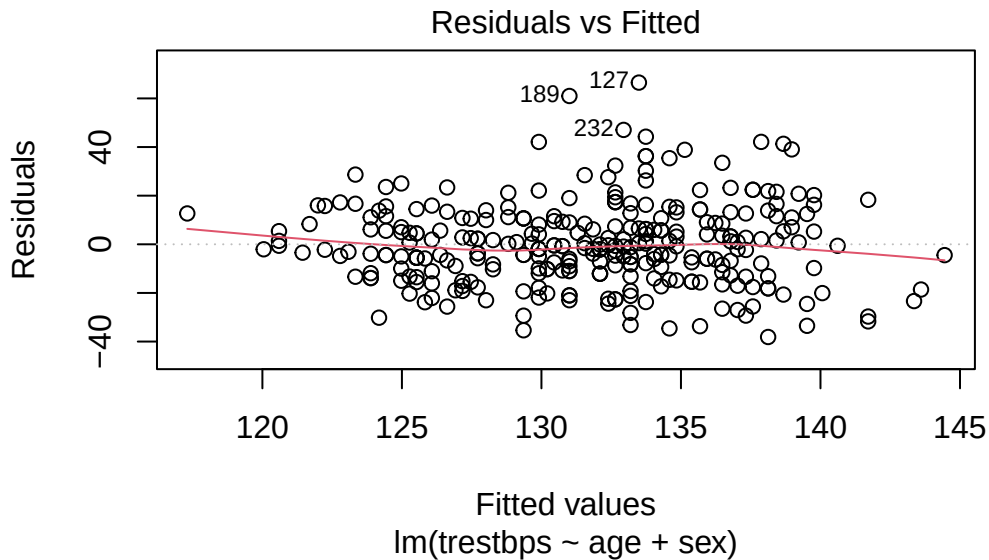
After adjusting for age, males have an estimated resting blood pressure that is **1.39 mm Hg lower** than females.

The 95% confidence interval for the male-female adjusted difference is **-5.51 to 2.72 mm Hg**, and the p-value is **0.506**.

There is no evidence of a statistically significant difference in resting blood pressure between males and females after adjusting for age.

A4. Model assumptions and interaction





The Q-Q plot should be checked for large departures from normality. In this dataset, the residuals are mostly compatible with approximate normality, although some tail deviations may be present.

The residuals-versus-fitted plot should be checked for curvature and unequal spread. There is no strong pattern suggesting that the additive linear model is inappropriate, although the model is not expected to explain all variation in resting blood pressure.

Anova Table (Type III tests)

Response: trestbps

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	71768	1	250.4720	< 2.2e-16 ***
age	7260	1	25.3374	8.328e-07 ***
sex	104	1	0.3633	0.5471
age:sex	149	1	0.5216	0.4707
Residuals	85673	299		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Answer

The age-by-sex interaction has $p = 0.471$. This does not provide evidence that the association between age and resting blood pressure differs between males and females. Together with the diagnostic plots, this suggests that the additive ANCOVA model is a reasonable model for these data.

Part B: Logistic regression and causal adjustment

B1. Preparing the outcome and exposure

	0	1
Heart Disease	0	139
No Heart Disease	164	0

The outcome is coded so that 1 is the event of interest: heart disease. The value 0 corresponds to no heart disease.

B2. Unadjusted model

Call:

```
glm(formula = target01 ~ chol, family = binomial, data = heart_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.989822	0.572848	-1.728	0.084 .
chol	0.003339	0.002271	1.470	0.142

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 417.98 on 302 degrees of freedom
Residual deviance: 415.78 on 301 degrees of freedom
AIC: 419.78

Number of Fisher Scoring iterations: 4

	OR	2.5 %	97.5 %
(Intercept)	0.3716428	0.1209252	1.142180
chol	1.0033444	0.9988878	1.007821

	OR	2.5 %	97.5 %
(Intercept)	0.3716428	0.1209252	1.14218
chol10	1.0339523	0.9889339	1.08102

Answer

In the unadjusted model, the odds ratio for a **1 mg/dl** increase in cholesterol is **1.003** with 95% CI **0.999 to 1.008** and $p = 0.142$.

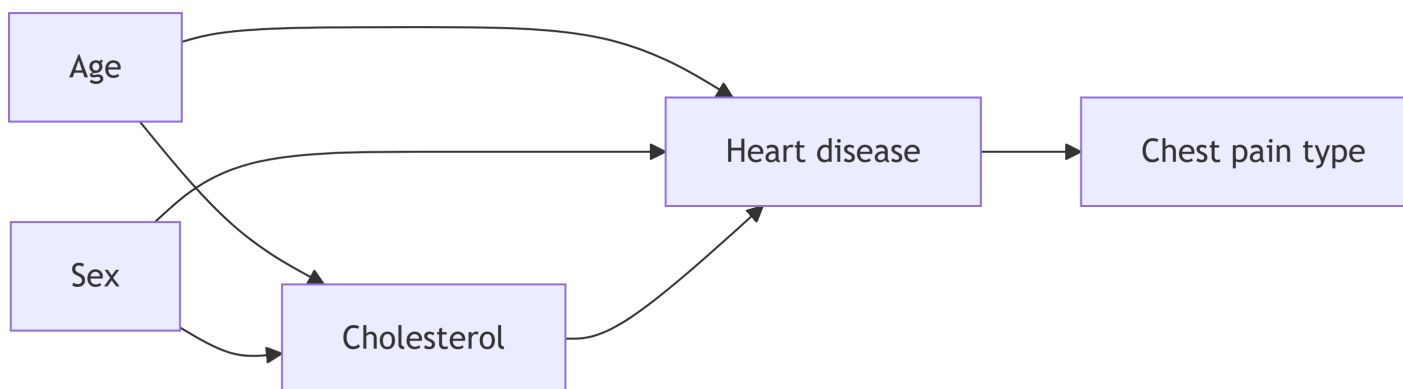
For a **10 mg/dl** increase in cholesterol, the odds ratio is **1.034** with 95% CI **0.989 to 1.081**.

The estimate is slightly above 1, but the confidence interval includes 1 and the p-value is not statistically significant. There is no clear evidence from the unadjusted model that cholesterol is associated with heart disease status.

B3. Causal diagram

The DAG contains the following arrows:

- age -> cholesterol
- age -> heart disease
- sex -> cholesterol
- sex -> heart disease
- cholesterol -> heart disease
- heart disease -> chest pain type



Answer

To estimate the causal effect of cholesterol on heart disease, the adjusted model should include **age** and **sex**.

Age and sex are confounders because they are causes of both cholesterol and heart disease. Chest pain type should not be adjusted for, because it is a consequence of heart disease in the DAG. Adjusting for a downstream consequence of the outcome is not part of the confounder adjustment set.

B4. Adjusted model

Call:

```
glm(formula = target01 ~ chol + age + sex, family = binomial,  
     data = heart_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-5.909350	1.095393	-5.395	6.86e-08 ***
chol	0.004829	0.002539	1.902	0.0572 .
age	0.062246	0.015120	4.117	3.84e-05 ***
sexMale	1.631814	0.303194	5.382	7.36e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 417.98 on 302 degrees of freedom
Residual deviance: 368.59 on 299 degrees of freedom
AIC: 376.59

Number of Fisher Scoring iterations: 3

	OR	2.5 %	97.5 %
(Intercept)	0.002713949	0.0003171035	0.02322749
chol	1.004841016	0.9998524987	1.00985442
age	1.064224022	1.0331489166	1.09623381
sexMale	5.113139619	2.8223350453	9.26332145

	OR	2.5 %	97.5 %
(Intercept)	0.002713949	0.0003171035	0.02322749

```
chol10      1.049478480 0.9985259652 1.10303099
age         1.064224022 1.0331489166 1.09623381
sexMale     5.113139619 2.8223350453 9.26332145
```

Answer

After adjustment for age and sex, the odds ratio for a **1 mg/dl** increase in cholesterol is **1.005** with 95% CI **1.000 to 1.010** and $p = 0.057$.

For a **10 mg/dl** increase in cholesterol, the adjusted odds ratio is **1.049** with 95% CI **0.999 to 1.103**.

The adjusted estimate is slightly larger than the unadjusted estimate. This suggests that age and sex act as confounders in the cholesterol-heart disease relationship, although the adjusted cholesterol association is still only borderline and is not statistically significant at the 0.05 level.

Part C: Building a prediction model

C1. Candidate model

Call:

```
lm(formula = thalach ~ age + sex + cp + chol + fbs + trestbps,
    data = heart_data)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-56.595 -11.437   2.564  13.335  43.999
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 187.98169   11.26288  16.690 < 2e-16 ***
age          -0.97352    0.13553  -7.183 5.61e-12 ***
sex1          1.02288    1.26254   0.810  0.4185
cp1           3.87332    3.29957   1.174  0.2414
cp2           6.33685    2.46981   2.566  0.0108 *
cp3           1.30012    2.09375   0.621  0.5351
chol          0.03689    0.02292   1.610  0.1086
fbs1         -0.46173    1.63623  -0.282  0.7780
trestbps      0.07576    0.06904   1.097  0.2734
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.71 on 294 degrees of freedom
 Multiple R-squared: 0.2775, Adjusted R-squared: 0.2578
 F-statistic: 14.11 on 8 and 294 DF, p-value: < 2.2e-16

Anova Table (Type III tests)

```
Response: thalach
      Sum Sq Df F value    Pr(>F)
(Intercept) 108185 1 278.5688 < 2.2e-16 ***
age          20039 1  51.5998 5.612e-12 ***
sex           255  1   0.6564  0.4185
cp           16642 3  14.2838 1.030e-08 ***
chol          1006 1   2.5906  0.1086
fbs           31  1   0.0796  0.7780
trestbps       468 1   1.2041  0.2734
Residuals    114178 294
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

For categorical predictors with more than two categories, such as `cp`, the Type III ANOVA table tests the predictor as a whole rather than only testing one dummy coefficient.

C2. Model selection

For this prediction exercise, backward elimination was used. The full candidate model was fitted first, and then the predictor with the largest p-value was removed as long as its p-value was greater than 0.10. This is reasonable for a prediction model because the goal is to obtain a more parsimonious model that keeps predictors with useful predictive information.

```
# A tibble: 3 x 3
  step removed p_value
<dbl> <chr>    <dbl>
1     1 fbs      0.778
2     2 sex      0.429
3     3 trestbps 0.235

[1] "age" "cp" "chol"
```

Answer

Backward elimination removed the following predictors:

1. `fbs`, $p = 0.778$
2. `sex`, $p = 0.429$
3. `trestbps`, $p = 0.235$

The final prediction model retained **age**, **chest pain type (cp)**, and **cholesterol (chol)**.

Call:

```
lm(formula = thalach ~ age + cp + chol, data = heart_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-58.701	-11.489	2.667	13.802	43.475

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	197.45207	9.05574	21.804	< 2e-16 ***
age	-0.92273	0.13019	-7.088	9.97e-12 ***
cpAtypical angina	1.97796	4.99947	0.396	0.692660
cpNon-anginal pain	-2.84347	4.63234	-0.614	0.539797
cpAsymptomatic	-15.97952	4.42906	-3.608	0.000362 ***
chol	0.04241	0.02242	1.891	0.059539 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.68 on 297 degrees of freedom

Multiple R-squared: 0.2723, Adjusted R-squared: 0.26

F-statistic: 22.22 on 5 and 297 DF, p-value: < 2.2e-16

	2.5 %	97.5 %
(Intercept)	179.630527922	215.27360552
age	-1.178934370	-0.66652072
cpAtypical angina	-7.860912318	11.81682362
cpNon-anginal pain	-11.959841240	6.27289858
cpAsymptomatic	-24.695840637	-7.26320698
chol	-0.001716395	0.08653757

The final fitted model is:

$$\widehat{\text{thalach}} = 197.45 - 0.923 \times \text{age} + \text{chest pain type coefficient} + 0.042 \times \text{cholesterol}.$$

With **Typical angina** as the reference category for chest pain type, the estimated chest-pain coefficients are approximately:

- Atypical angina: 1.98
- Non-anginal pain: -2.84
- Asymptomatic: -15.98

C3. Prediction for a patient

The patient has:

- age: 62 years
- sex: female
- chest pain type: typical angina
- cholesterol: 240 mg/dl
- fasting blood sugar > 120 mg/dl: false
- resting blood pressure: 135 mm Hg

Only age, chest pain type, and cholesterol are used, because these are the predictors retained in the final model.

1
150.4215

Answer

The predicted maximum heart rate for this patient is **150.4 beats per minute**. This is calculated from the final model as:

$$197.45 - 0.923 \times 62 + 0.042 \times 240 = 150.4.$$